Sadegh Ebrahimi

Machine learning, electrical and neural engineer/scientist

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SUMMARY

- Electrical engineer and ML scientist who builds real-time ML systems end to end: sensors and embedded firmware, signal processing, models, cloud pipelines, and LLM products.
- 6+ years across the BCI stack: recording, signal processing, decoding, stimulation, embedded deployment.
- Shipped real-time decoders for an implanted BCI used by people with paralysis; 15+ trial patients.
- Co-founded Sylk AI, a live LLM system that turns breaking financial news into stock-move predictions and scores every call against the market.
- Leads cross-functional teams: directed a team of 7; led about 10 engineers to ship a wearable medical device.
- PhD in Electrical Engineering, Stanford; first-author *Nature*; NeurIPS; co-inventor on U.S. patents.

EDUCATION

Stanford University, PhD, MSc Electrical Engineering (GPA 3.9) *2012 – 2019*

Sharif University of Technology, BSc Electrical Engineering (GPA 3.9, top of class) *2008 – 2012*

EXPERIENCE

Co-Founder · Sylk AI *2025 – Present*

- Built, end to end, a live system that turns SEC filings and news into stock-move predictions every 5 minutes: Python pipelines, PostgreSQL, a React 19/Next.js front end, and monitoring.
- Designed a multi-stage LLM pipeline on open-weight models with structured outputs, validation guardrails, and quota-aware failover; magnitudes are calibrated to each stock's realized volatility.
- Scored every prediction against real prices; error analysis led to a two-stage filter that raised accuracy and cut token usage 70%.
- Fine-tuned Llama 3 8B with LoRA on about 40,000 financial articles paired with stock outcomes (AWS EC2), raising prediction accuracy 18%.
- Added retrieval (RAG) over a moving window of recent news, so each article is judged in the context of the events unfolding around it.

Senior ML Engineer · Synchron *Aug 2022 – Sep 2026*

- Developed real-time cursor-control and multi-switch decoders (ML models) for an implanted BCI; participants with paralysis achieved 2D cursor and multi-switch control.
- Engineered a drift-robust neural feature for brain activity decoders that held performance for weeks.
- Built transformer foundation models of neural ECoG data that generalize to new users via domain-adversarial learning; deployed decoders on embedded systems with ONNX.
- Directed a team of 7 (3 BCI scientists and 4 field clinical engineers) through a multi-site clinical study.
- Used Kalman filtering for real-time cursor control; deployed decoders with ONNX on ARM Cortex-M4 microcontrollers.
- Benchmarked foundation models against RNN/LSTM, XGBoost, SVM, and LDA decoders.
- Cut system startup time 95% with a personalization model.
- Launched the automated ML pipeline for SyncLab, Synchron's clinical data and ML platform, which trains and evaluates models on every new data collection, with dashboards that catch regressions.

- Traced signal anomalies in a new implant hardware revision to root causes with the EE team.

Lead Electrical Engineer · Neuvotion *Aug 2021 – Dec 2022*

- Led about 10 contract engineers across firmware, hardware, app, and cloud to ship NeuStim, a wearable neuromuscular stimulator that fuses EMG and IMU sensing with real-time on-device ML.
- Distilled a CNN gesture classifier onto an ESP32 under 10MB within TF-Lite operator limits, separating 7 trigger gestures; tuned for very low false positives under safety constraints.
- Co-invented a wearable patch with on-board motion sensing and addressable electrodes (patent pending).

Lead Electrical Engineer · Feinstein Institute for Medical Research *Aug 2021 – Aug 2022*

- Designed and built much of the initial real-time infrastructure for a bidirectional BCI: Utah-array decoding drove muscle stimulation; fingertip sensors drove cortical stimulation (later in *Nature Medicine*).
- Decoded hand-motor signals from a Utah array 6+ years after implant; analyzed hand EMG.
- Co-authored 2 clinical studies of spinal cord stimulation (*Frontiers in Neuroscience*).

Postdoctoral Scholar & PhD Researcher · Stanford University *Sep 2012 – Aug 2021*

- PhD (Schnitzer Lab): designed and led a large-scale two-photon imaging study of more than 21,000 neurons across eight visual cortical areas, tracked over days, from experiment design to the preprocessing, analysis, and decoding pipelines.
- Showed how reliable coding and inter-area communication emerge despite day-to-day variability in individual neurons (first-author *Nature*, 2022, with a *Nature* News & Views commentary).
- Postdoc (Ganguli Lab): built a 270-degree immersive VR system for in-vivo imaging and GAN-based models that reconstruct viewed scenes from cortical activity; won a Simons Foundation grant.
- Co-authored a geometric and dynamical theory of latent computation in neural networks (under review, *Nature Neuroscience*) and ActSort, active-learning cell sorting for large-scale calcium imaging (NeurIPS 2024).

Software Engineer Intern · Oracle *Jun 2017 – Sep 2017*

- Designed a real-time clustering algorithm for network failure events (U.S. Patent 10,678,610).

SKILLS

Machine learning: Transformers, CNNs, RNNs/LSTMs, GANs; domain adaptation, model compression and distillation, time-series modeling, Bayesian state estimation (Kalman filtering), evaluation design, reinforcement learning (RLVR).

LLMs: Open-weight LLMs (Llama, GPT-OSS) in production; LoRA fine-tuning, RAG, structured outputs, validation guardrails, multi-stage pipelines, LLM evaluation, cost optimization.

Neurotech & signals: Neural decoding and encoding, ECoG, Utah arrays, EMG, IMU, two-photon imaging, closed-loop and spinal cord stimulation, signal processing, clinical studies.

Embedded: Embedded C/C++, ARM Cortex-M4, ESP32, real-time firmware, on-device inference (ONNX, TF-Lite), sensor interfaces, stimulation hardware.

Software: Python, PyTorch, JAX, TensorFlow, scikit-learn, CUDA, C/C++, TypeScript, React 19/Next.js, PostgreSQL/SQL, AWS, Docker, CI/CD, Linux, Git.

HONORS & RECOGNITION

- Simons Foundation grant for reconstructing viewed scenes from brain activity (Stanford postdoc).
- Ranked 24th of about 300,000 applicants on Iran's national university entrance exam; top of the electrical engineering class at Sharif University of Technology.
- PhD research featured in *Nature* News & Views, HHMI, Stanford HAI, and Neuroscience News; clinical trials at Northwell Health covered by CBS New York.

PUBLICATIONS & PATENTS

Journal and Conference Papers

1. **Ebrahimi, S.**, Lecoq, J., Rumyantsev, O., Tasci, T., Zhang, Y., Irimia, C., Li, J., Ganguli, S. and Schnitzer, M.J. (2022). *Emergent reliability in sensory cortical coding and inter-area communication*. **Nature** 605, 713–721.
2. Jiang, Y., Akengin, H.O., Zhou, J., Aslihak, M.A., Li, Y., Chrapkiewicz, R., Hernandez, O., **Ebrahimi, S.**, et al. (2024). *ActSort: An active-learning accelerated cell sorting algorithm for large-scale calcium imaging datasets*. **Advances in Neural Information Processing Systems (NeurIPS)** 37, 79095–79143.
3. Chandrasekaran, S., Bhagat, N.A., Ramdeo, R., **Ebrahimi, S.**, Sharma, P.D., Griffin, D.G., Stein, A., Harkema, S.J. and Bouton, C.E. (2023). *Targeted transcutaneous spinal cord stimulation promotes persistent recovery of upper limb strength and tactile sensation in spinal cord injury: a pilot study*. **Frontiers in Neuroscience** 17, 1210328.
4. Chandrasekaran, S., Bhagat, N.A., Ramdeo, R., **Ebrahimi, S.**, Sharma, P.D., Griffin, D.G., Stein, A., Harkema, S.J. and Bouton, C.E. (2023). *Case study: persistent recovery of hand movement and tactile sensation in peripheral nerve injury using targeted transcutaneous spinal cord stimulation*. **Frontiers in Neuroscience** 17, 1210544.
5. **Ebrahimi, M.S.** and Abadi, H.K. (2021). *Study of residual networks for image recognition*. In **Intelligent Computing**: Proceedings of the 2021 Computing Conference (pp. 754–763). Springer.
6. **Ebrahimi, M.S.**, Hills, G., Sabry, M.M., Shulaker, M.M., Wei, H., Wu, T.F., Mitra, S. and Wong, H.S.P. (2014). *Monolithic 3D integration advances and challenges: From technology to system levels*. In **IEEE S3S Conference** (pp. 1–2).
7. **Ebrahimi, M.S.**, Daraei, M.H., Behzadan, V., Khajooeizadeh, A., Behrostaghi, S.A. and Tajvidi, M. (2011). *A novel utilization of cluster-tree wireless sensor networks for situation awareness in smart grids*. In **IEEE PES Innovative Smart Grid Technologies** (pp. 1–5).

Preprints

1. Dinc, F., Blanco-Pozo, M., Klindt, D., Acosta, F., Sylber, C., Jiang, Y., **Ebrahimi, S.**, Shai, A., Tanaka, H., Yuan, P., Miolane, N. and Schnitzer, M.J. (2026). *A geometric and dynamical theory of latent computations in biological neural networks*. **bioRxiv** (under review at **Nature Neuroscience**).
2. Chetty, N., Kacker, K., Feldman, A.K., Yoo, P.E., Bennett, J., Fry, A., Tal, I., Hardy, N.F., **Ebrahimi, S.**, et al. (2025). *Signal properties and stability of a chronically implanted endovascular brain computer interface*. **medRxiv**.
3. Dinc, F., Blanco-Pozo, M., Klindt, D., Acosta, F., Jiang, Y., **Ebrahimi, S.**, Shai, A., Tanaka, H., Yuan, P., Schnitzer, M.J. and Miolane, N. (2025). *Latent computing by biological neural networks: A dynamical systems framework*. **arXiv**:2502.14337.
4. Chandrasekaran, S., Bhagat, N.A., Ramdeo, R., **Ebrahimi, S.**, Sharma, P.D., Stein, A., Griffin, D.G., Harkema, S.J. and Bouton, C.E. (2022). *Targeted transcutaneous cervical spinal cord stimulation promotes upper limb recovery in spinal cord and peripheral nerve injury*. **medRxiv**.

Patents

1. Hardy, N.F., Echavarria, C., Herbers, C., **Ebrahimi, S.**, Bennett, J. and Yoo, P.E. (2025). *Systems for controlling one or more devices using a signal control unit*. **U.S. Patent 12,461,596**.
2. Hardy, N.F., Echavarria, C., Herbers, C., **Ebrahimi, S.**, Bennett, J. and Yoo, P.E. (2026). *Systems for controlling one or more devices using a signal control unit*. **U.S. Patent Application 19/341,581**.
3. Bouton, C., **Ebrahimi, S.**, Toro, J., Leavitt, J. and Niewiadomski, J. (2025). *Addressable serial electrode arrays for neurostimulation and/or recording applications and wearable patch system with on-board motion sensing and magnetically attached disposable for rehabilitation and physical therapy applications*. **U.S. Patent Application 18/854,837**.
4. **Ebrahimi, M.S.**, Patti, R.H.R. and Garvey, D. (2020). *Using and updating topological relationships amongst a set of nodes in event clustering*. **U.S. Patent 10,678,610** (Oracle).